

10. (a) Dave says the distances of each leg in a long-distance triathlon are all double those in a middle-distance triathlon.

Tom says the distances of each leg in an Olympic triathlon are double those in a sprint triathlon.

Use data from **Table 1** to explain whether Dave or Tom is correct.

[2]

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- (b) Piera says Jake breathed in less air per minute at 4 minutes than at 3 minutes because his breathing rate was lower.

Petula disagrees because Jake's airflow per breath was greater at 4 minutes than at 3 minutes.

Use data from **Table 2**, and the equation below it, to explain whether you agree with Piera or Petula.

[2]

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- (c) Use the information in **Table 4** and equations from pages 4 and 5 to answer the following questions.

- (i) Calculate Malcolm's BMI.

[2]

BMI =

- (ii) Calculate Malcolm's maximum heart rate.

[1]

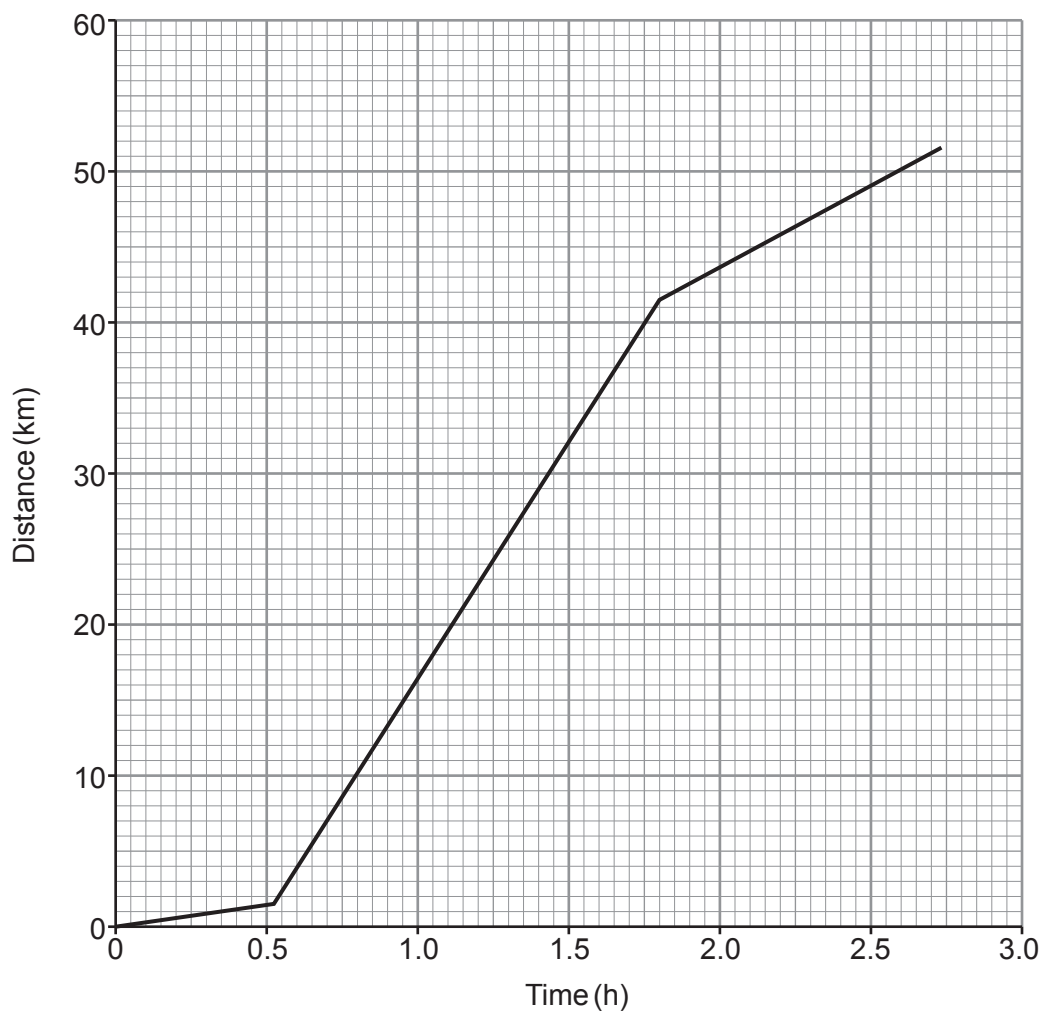
maximum heart rate = bpm



- (d) Karen wants to exercise in the aerobic zone. Use the information in **Tables 3 and 4** and on page 4 to determine the range in heart rate she should aim for. [1]

heart rate range is from bpm to bpm.

- (e) Data in the **Triathlon times** section (page 5) and **Table 1** was used to plot distance-time graphs for each of the triathlons described. There were three sections in each graph to show the swim, cycle and run legs. The graphs did not include transition times. One of the graphs is shown below.



Determine which type of triathlon is represented by the graph and give **two** reasons for your answer. [3]

Type of triathlon:

Reason 1:

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Reason 2:

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- (f) Use the information about Triathlon times on page 5, the information in **Table 1** and an equation on page 7 to calculate the mean speed during an Ironman triathlon. [3]

mean speed = km/h

- (g) When a triathlete mounts their bike they are travelling at 1.5 m/s. Use the information in **Table 5** and an equation on page 7 to calculate the acceleration along level ground in a cycle leg. [3]

acceleration = m/s²

- (h) It is thought that the higher the age group of the triathlete, the lower the mean speed in the cycle leg. Use data from **Graph 1** to explain whether this statement is true for all age groups. [2]

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END OF PAPER



	Group
1	1
2	2
3	3
4	4
5	5
6	6
7	7
0	0

1 H Hydrogen 1 4 He Helium 2																			
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10		
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	
										Key									

Key

- relative atomic mass

Ar
Symbol
Name
Z
atomic number